

Respiratory Presentations of Exotic Companion Mammals

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Learning Outcomes

1. Compare the clinical presentation and available diagnostics of the most common respiratory diseases of ferrets, rabbits and rats.
2. Apply prognostic knowledge of the most common respiratory diseases of ferrets, rabbits and rats to clinical cases.
3. Select appropriate antimicrobial therapies for exotic companion mammals suffering from respiratory infections.

While respiratory diseases present in all exotic companion mammals, the three most affected in clinical practice are ferrets, rabbits and rats. Clinical signs of upper and lower respiratory pathology that are noted in traditional companion animals can be broadly applied to the three species we will focus on for this lecture with a few notable differences:

- As obligate nasal breathers, thick nasal discharge (purulent or hemorrhagic) can be very distressing for a rabbit and prominent debris can lead to open mouth breathing and decompensation quickly.
- The Harderian lacrimal glands of rats produces porphyrin laden secretions. While the normal rat is a fastidious groomer, sick rats may be grooming less, and red-brown material may accumulate around the eyes and nose. Clients may mistake this for blood. A quick trick in the exam room is that most porphyrins will fluoresce with Wood's lamp or black light.
- Ferrets have a large respiratory system and thus, have heightened compensatory capabilities. Thus, even in states of moderate radiographically evident lower respiratory disease a ferret may show only subtle or mild respiratory signs. Rabbits and rodents have much smaller thoracic space, and thus, less extensive pathology can lead to more severe clinical presentations.
- You may not see nasal discharge around the nares in fastidious groomers like rabbits until more advanced stages, look for dried discharge on the medial aspect of the antebrachia.

Amongst all mammals, the most common body system differential for primary respiratory disease is cardiovascular disease. Given this, chest radiographs have the highest diagnostic utility and should be a first choice in attempting to determine which body system is causing respiratory signs. Point of care thoracic ultrasound can also be used, requiring less sedation and aiding in detection of animals with lung lobe consolidation or pleural effusion. A less obvious differential for upper respiratory signs in species with hypselodont dentition (rabbits and rodents) is apical overgrowth that extends into the sinus or nasal cavities either due to advanced dental disease, odontogenic abscessation or elodontomas.

Ferrets

When diagnosed with respiratory disease, lower respiratory disease is far more common than upper respiratory disease. Overwhelmingly, pneumonia is the most common respiratory pathology encountered, and viral, bacterial or fungal etiologies should be suspected, with viral etiologies being the most important differential.

Viral pneumonias

Ddx: Canine distemper, influenza, COVID-19

History: Ruling in or out viral etiologies of pneumonia should start with a good history. Ferrets suffering from CDV would be unvaccinated (or unknown vaccination) and are often recently obtained from the pet store. Contact with an infected animal (usually a dog) is necessary but not always present in the history. Ferrets suffering from Influenza or COVID-19 can be screened by asking questions about similar signs of human caretakers.

Clinical Presentation: Severity of clinical signs is based upon the viral etiology. Ferrets afflicted with CDV may also suffer from dermatologic (erythematous and pruritic rashes that progress to hyperkeratosis and crusts on the footpads and face) and neurologic (paresis, convulsions, tremors) signs and their clinical signs are progressive. In comparison, infections with Influenza and COVID are often mild and self-limiting, except in older ferrets with comorbidities.

Diagnostics:

- Radiographs often show an interstitial pattern
- Serum/plasma chemistry is often unremarkable.
- CBC changes may include an inflammatory leukogram and anemia of chronic disease (CDV)
- PCR is available for detection of viral DNA after sampling nasal discharge.
- Serology to detect antibodies against CDV is available and can be useful to assess for prior vaccination in healthy animals or likely CDV in sick animals with no vaccination status.

Treatment (CDV): There are no successful treatment options for ferrets with CDV and the infection is almost uniformly fatal. While awaiting diagnostics, general supportive care can be provided (NSAIDS, nutritional support, tier 1 antibiotics to prevent secondary bacterial pneumonia).

Treatment (Influenza/COVID): Given that most cases are mild, only minimal supportive measures are required. In cases where ferrets are exhibiting prolonged and/or more severe clinical signs, human antiviral drugs can be considered with some literature available to guide clinicians on dosing.

Prognosis: Infection with CDV carries a grave prognosis. Infections with COVID or influenza are generally mild and self-limiting, except in rare cases.

Mycotic pneumonias

Several pulmonary mycoses have been reported in ferrets including *Blastomyces dermatitidis*, *Coccidioides immitis* and *Cryptococcus* spp. Clinicians should be suspicious of a mycotic pneumonia if they live in an endemic region and have systemic and/or distant signs of infection in addition to lower respiratory signs. The prognosis for ferrets with mycotic pneumonia is often poor.

Rabbits

When diagnosed with respiratory disease, upper respiratory disease is far more common than lower respiratory disease in rabbits. Overwhelmingly, rhinosinusitis, commonly coined “snuffles”,

is the most frequent respiratory pathology encountered, and bacterial causes are the most common.

Rhinosinusitis

Pertinent anatomy: Rabbits have paired symmetric paranasal sinuses abaxial to the nasal cavity. These large areas drain into the nasal cavity in a small slit, these anatomic findings paired with the caseous nature of rabbit inflammatory debris is a recipe for chronic, poorly draining infections. Further, the apices of the maxillary premolars and rostral molars if elongated, can pierce into the paranasal sinuses and be nidus for infection.

Clinical Presentation: Rabbits with rhinosinusitis will exhibit expected signs of upper respiratory infection including sneezing and nasal discharge. Extension of infection through the nasolacrimal duct may lead to ocular discharge and conjunctivitis while extension through the Eustachian tube can lead to middle ear infections. Seeding of bacteria down the trachea may lead to pneumonia. Rabbits rarely exhibit fever or systemic signs, except when infection has travelled systemically to other organs leading to hepatic, renal, or lung involvement.

Important differentials to consider:

- Rabbit syphilis, caused by *Treponema cuniculi*, can lead to crusting and exudate around the nares, which can be confused with nasal discharge.
- Myxomatosis can cause oculonasal swelling with some serous nasal discharge.
- Fungal infections of the sinuses are rare but may be encountered.
- Nasal mucosal adenocarcinoma is reported and precipitated a chronic rhinitis.

Bacteria involved (**these bacteria can be normal inhabitants of the upper respiratory tract)

- *Pasteurella multocida* **
- *Bordetella bronchiseptica* **
- *Pseudomonas spp.*
- *Staphylococcus spp.* **
- *E. coli*
- Mixed infections are common

Diagnosis:

- Complete blood cell counts and serum/plasma chemistry is often unremarkable.
- Super straight skull radiographs may facilitate in detecting asymmetries of the nasal and sinus passages, bony lyses and detect apical involvement.
- Skull CT is the most sensitive in evaluating the nasal and sinus spaces and detecting dental involvement.
- Collection of materials for culture or next-generation sequencing from the nares may or may not include the microbe that is causing the infection, as that microbe is primarily in the sinus and spilling out into the nasal cavity. Only deep nasal cultures should be utilized, which often necessitate heavy sedation.

Treatment: Surgical (pararhinotomy or dorsal rhinotomy), medical (long term antibiotics) therapy have may be pursued. Antibiotics that are toxic to hind gut fermenters should never be utilized and antimicrobial therapy is often required for 1-3 months for resolution of clinical signs. When

not guided by culture and sensitivity, empirical choices for antibiotics should start with tier 1 antimicrobials, aligning with antimicrobial stewardship.

Prognosis: Given the anatomy of the sinuses, thick nature of the inflammatory material present and advanced state that many rabbits present, clinically controlling infections is more reasonable than cutting them.

Rats

The number one reason for veterinary evaluation of rats is to evaluate for respiratory pathology. Rats present with both upper and lower respiratory signs when they present with abnormalities of this body system.

Chronic Respiratory Disease (CRD) of Rats

Pathophysiology: Multiple infectious agents, either present in a patient individually or more often, in concert, lead to CRD. Rats are infected at the pet store, when they encounter rats from multiple different sources or at the breeding facility, where biosecurity is low.

Infectious etiologies involved (**major respiratory pathogens in this complex)

- *Mycoplasma pulmonis* **
- *Streptococcus pneumonia* **
- *Corynebacterium kutscheri*.
- Sendai virus (paramyxovirus)
- *Pneumocystis carinii*
- *Filobacterium rodentium* (ciliary associated respiratory bacillus)
- *Haemophilus* spp.

Clinical Presentation: Presenting rats may be mildly affected (sneezing, increased nasal discharge) to presenting in respiratory distress that is poorly responsive to oxygen therapy. Given the multiple infectious agents that can be at play, this may be expected. Mycoplasmosis is endemic in pet rats and the pathogen by itself causes mild waxing and waning of clinical signs and development of a chronic bronchitis. However, microscopic damage to the respiratory tract can predispose to invasion with other agents that can cause significant disease. Infection with *Streptococcus* and *Corynebacterium* are more likely to cause severe purulent pneumonias with far more severe respiratory signs noted.

Differential Diagnoses:

- Neoplasia is common in pet rats and they can suffer from primary lung neoplasia or metastatic lesions.
- Cardiac dysfunction

Diagnosis:

- Radiographs can aid in screening for more severe pneumonia which would point to *Streptococcus*, *Corynebacterium* or another secondary invader and can also identify cardiomegaly and masses that may be more consistent with neoplasia. Rats suffering from Mycoplasmosis may not have any appreciable pulmonary radiographic changes.

- Thoracic CT is more sensitive for lung changes than radiographs and can detect bronchitis lesions caused by Mycoplasmosis.
- Thoracic point of care ultrasound can be used, typically without sedation, to identify fluid and consolidated lung tissue.
- Aspiration of lung lesions that are safe to access can aid in the clinical evaluation of whether a pulmonary lesion is neoplastic or inflammatory through cytology and culture.
- Serology for *Mycoplasma* and Sendai virus are available, using a small amount of blood through Charles River Laboratories.

Treatment: Mildly affected rats that are thought to be suffering from Mycoplasmosis without a secondary invader can be managed with bronchodilators and intermittent use of antimicrobials with mycoplasmocidal action (tetracyclines, macrolides). Rats with significant respiratory compromise and radiographs evidence of pneumonia require more aggressive therapy with antimicrobials effective against gram positive bacteria (potentiated amoxicillins, second and third generation cephalosporins) along with other supportive measures (oxygen therapy, anti-inflammatories, bronchodilators).

Prognosis: Rats can live for several years with CRD secondary to Mycoplasmosis, with waxing and waning signs of their bronchopneumonia. Secondary infections can be severe and life limiting, even causing sudden death in young animals <6-9 months of age.